WEEK 3.SESSION 1

Enterprise Data Engineering, Analytics & AI

Power BI Desktop Tour, Data Sources & Importing from Excel / CSV

Foundational Module 3 — Power BI Foundations & Basics

Corporate Training Program | Trainer Name: Prachi Kabra | Hands-On Learning

Power BI Desktop

- ✓ Interface Tour
- ✓ Data Sources
- ✓ Excel / CSV Import

SESSION 1 — AGENDA

01 | **Power BI Ecosystem Overview** | **10 min**

What is Power BI? Desktop vs Service vs Mobile. Licensing overview.

02 | **Power BI Desktop Interface Tour** | **20 min**

Ribbon, Report/Data/Model views, Fields & Filters panes, Canvas.

03 | **Understanding Data Sources** | **15 min**

Supported source types: files, databases, cloud, web, APIs.

04 | **Importing Data from Excel** | **20 min**

Selecting sheets/tables, column types, handling headers.

05 | **Importing Data from CSV** | **15 min**

Delimiters, encoding, data type detection, common issues.

06 | **Hands-On Practice** | **10 min**

Live import exercise + Q&A

POWER BI ECOSYSTEM OVERVIEW

Power BI Desktop

- Free to download & use
- Main authoring tool
- Connect, transform & visualize
- Windows application (.exe)
- Publish to Power BI Service

Power BI Service

- Web-based (app.powerbi.com)
- Share & collaborate on reports
- Schedule data refresh
- Dashboards & workspaces
- Requires Pro/Premium license

Power BI Mobile

- iOS and Android apps
- View published reports
- Offline access to dashboards
- Alerts & notifications
- Optimized for touch

💡 This session focuses on Power BI Desktop — the tool you will use daily for data connection and report authoring.

POWER BI DESKTOP — INTERFACE TOUR

① RIBBON — Home, Insert, Modeling, View, Optimize, Help tabs

② VIEWS Report Data Model

REPORT CANVAS
(Design your visuals here)

③ FILTERS Pane

④ FIELDS Pane

⑤ PAGE TABS — Report Pages Navigation

Ribbon shortcut: ALT key reveals tab shortcuts

Switch views using left-side icons (Report / Data / Model)

Right-click Fields pane items for more options

THE THREE VIEWS EXPLAINED

REPORT VIEW

- Your main workspace for building reports
- Drag fields onto the canvas to create visuals
- Add slicers, cards, charts, tables & maps
- Manage pages like Excel worksheets
- Use Format pane to style each visual
- Switch between Selection & Filters pane

DATA VIEW

- Browse your loaded tables row-by-row
- Verify data was imported correctly
- See calculated columns you've created
- Check data types assigned to columns
- Cannot edit raw data here — use Power Query
- Search for specific column names quickly

MODEL VIEW

- See all tables and their relationships visually
- Drag between columns to create relationships
- Set relationship cardinality (1:many, many:many)
- Manage cross-filter direction
- Hide fields not needed in reports
- Foundation for accurate DAX measures

UNDERSTANDING DATA SOURCES IN POWER BI

How to Get Data: Home → Get Data (or use the Get Data button on the canvas)

File Sources

- Excel Workbook (.xlsx, .xls)
- CSV / Text files
- XML & JSON
- PDF tables
- Folder (batch import)

Database Sources

- SQL Server
- MySQL / PostgreSQL
- Oracle Database
- Azure SQL / Synapse
- Access Database

Cloud & Online

- SharePoint List / Folder
- Azure Blob Storage
- Dynamics 365
- Google Analytics
- Salesforce Objects

Other Sources

- Web (scrape tables)
- OData Feed
- REST API (blank query)
- Python / R Script
- ODBC / OLEDB

IMPORTING DATA FROM EXCEL — STEP BY STEP

1

Get Data → Excel Workbook

Home ribbon → Get Data → Excel Workbook. Browse and select your .xlsx / .xls file. Power BI supports Excel 97-2003 (.xls) onwards.

2

Navigator — Select Sheets or Tables

Navigator window shows all Sheets and named Tables. ✓ Always prefer named Tables over raw Sheets for reliable column headers and auto-expansion.

3

Preview & Load vs Transform

Click Load to import as-is. Click Transform Data to open Power Query Editor first (recommended). Review data types before loading.

4

Check Column Data Types

Power BI auto-detects types (text, number, date). Verify in the Data View. Wrong types cause chart/filter issues. Fix in Power Query.

5

Refresh & Connection

File path is stored. When source file changes, click Refresh. Use Transform Data → Data Source Settings to update the path.

EXCEL IMPORT — BEST PRACTICES & COMMON ISSUES

✅ BEST PRACTICES

- Structure data as an Excel Table (Ctrl+T) before importing
- Use clear, single-row headers — no merged cells
- Keep one dataset per sheet / table
- Remove blank rows and columns within data
- Store dates in ISO format: YYYY-MM-DD
- Avoid formulas that reference other workbooks
- Name your tables meaningfully (e.g. 'Sales_2024')

⚠️ COMMON ISSUES & FIXES

⚠ Dates imported as text

→ Change column type to Date in Power Query

⚠ Extra header rows / blank rows

→ Use 'Use First Row as Headers' in Power Query

⚠ Numbers with currency symbols

→ Remove symbols, change type to Decimal

⚠ Merged cells causing nulls

→ Un-merge before importing or fill down in PQ

⚠ Sheet name changed = error

→ Use a named Table instead of sheet reference

IMPORTING DATA FROM CSV FILES

CSV (Comma-Separated Values) — Plain text files where each line is a row and values are separated by a delimiter (comma, semicolon, tab, pipe)

IMPORT STEPS

1. Home → Get Data → Text/CSV
2. Browse and select your .csv or .txt file
3. Preview window shows detected delimiter & encoding
4. Change delimiter if incorrect (File Origin = 65001 UTF-8)
5. Click Transform Data to shape, or Load to import directly
6. Set correct data types in Power Query before closing

KEY SETTINGS TO CHECK

Delimiter

Comma, Semicolon, Tab, Pipe, Custom

File Origin

65001: UTF-8 (most common)
1252: Western Europe
932: Japanese (Shift-JIS)

Data Type Detection

Based on first 200 rows
Always verify — change if needed

COMMON CSV PITFALLS

- 🔴 Wrong delimiter → one column with all data merged
- 🔴 Encoding mismatch → garbled special characters (é, ñ, ü)
- 🔴 Quoted fields with commas inside → parsing errors
- 🔴 No headers → Power BI creates Column1, Column2... rename manually

EXCEL vs CSV — WHEN TO USE WHICH?

Feature	Excel (.xlsx)	CSV (.csv)
Multiple sheets/tables	✅ Yes	❌ No (single flat table)
Data types preserved	✅ Yes (date, number, text)	⚠️ Partial (all text initially)
File size for large data	⚠️ Heavier (.xlsx format)	✅ Lighter (plain text)
Interoperability	⚠️ Requires Excel/Office	✅ Any software can read
Formulas / Formatting	✅ Stored in file	❌ Not stored in CSV
System exports (ERP, CRM)	⚠️ Sometimes available	✅ Most common export format
Power Query: named tables	✅ Preferred method	❌ No table concept
Best for	Internal structured reports	Data exports from systems

💡 Pro Tip: When given a choice, request data as Excel with named Tables. If receiving system exports, expect CSV — always check delimiter and encoding.

QUICK REFERENCE — KEYBOARD SHORTCUTS & TIPS

KEYBOARD SHORTCUTS

Ctrl + Z	Undo last action
Ctrl + M	New measure (DAX)
Alt + F4	Close Power BI Desktop
Ctrl + Shift + F	Full-screen focus mode
Ctrl + S	Save the report
Ctrl + D	Duplicate visual
F5	Refresh data preview
Ctrl + Enter	Apply Power Query step

SESSION 1 KEY TAKEAWAYS

- 🖥️ Power BI Desktop is free — the primary tool for building reports
- 📊 3 Views: Report (design), Data (inspect rows), Model (relationships)
- 📁 Get Data → 150+ connectors — File, Database, Cloud, Web & more
- ✅ Excel: prefer named Tables; CSV: always check delimiter & encoding
- ⚡ Transform Data opens Power Query — always review types before Load

HANDS-ON EXERCISE

Practice what you've learned — follow along!

Task 1

Excel Import

1. Download the sample Sales_Data.xlsx file
2. Home → Get Data → Excel Workbook
3. In Navigator, select the 'Sales_Table' table
4. Click Transform Data — review column types
5. Close & Apply — verify in Data View

Task 2

CSV Import

1. Download the sample Customers.csv file
2. Home → Get Data → Text/CSV
3. Check delimiter (should be Comma)
4. Check File Origin (should be 65001: UTF-8)
5. Load — then verify in Data View

Task 3

Interface Exploration

1. Switch between Report, Data and Model views
2. In Data View — find column with wrong data type
3. In Model View — observe the table relationship
4. In Report View — drag a field to the canvas
5. Delete the visual (Backspace)

❓

Questions & Discussion

Week 3 Session 1 Complete

Session Recap:

- **Power BI Desktop Tour**
- Data Sources & Get Data
- Excel Import
- CSV Import

Next Session → S2: Power Query Editor — Transformations, Data Shaping & Cleaning